

A PROGRESSIVE QUADRATIC FAILURE CRITERION FOR A LAMINATE*

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Abstract

The application of failure criteria to laminated composites is a critical step in the determination of the strength and safety of a point in a structure under combined loading conditions. Non-homogeneous stresses within a structure may induce a complicated failure scenario whereby one ply at a point can initiate failure and affect other plies at the same point or the same ply at a neighboring point. After the first-ply failure, the stiffness of the ply is reduced by either matrix or fiber failures. The strength of the laminate at the same point is evaluated again to see if the laminate can carry an additional load. This ply-by-ply analysis progresses until the ultimate strength of the laminate is reached. The process can be extended to structures subjected to non-homogeneous stresses where both ply-by-ply and point-to-point strength analyses are progressively applied. © 1998 Elsevier Science Ltd. All rights reserved

Keywords: failure, progressive, plotting envelopes

NOTATION

E_i	Young's modulus in the i th direction; $i = x, y, s$
E_i^0	In-plane Young's moduli of a laminate; $i = 1, 2, 6$
E_f	Fiber longitudinal Young's modulus
E_f^*	Fiber degradation factor (to model fiber collapse)
E_x, E_y	Longitudinal and transverse Young's modulus of a ply
E_m	Matrix Young's modulus
E_m^*	Matrix degradation factor (to model a ply with micro-cracks)
E_s	Shear modulus of a unidirectional ply
F_{ij}, F_i	Strength parameters in stress space of a quadratic failure criterion; $i = x, y, s$

F_{xy}^*	Normalized interaction term of a quadratic failure criterion
FPF	First Ply Failure
G_{ij}, G_i	Strength parameters in strain space of a quadratic failure criterion; $i = x, y, s$
h_0	Unit ply thickness
k	Failure index, equal to $1/R$
LPF	Last Ply Failure
$\{N\}, N_i$	Absolute in-plane stress resultant components; $i = 1, 2, 6$
n	Exponent for longitudinal compressive strength degradation
R	Strength/stress ratio or strength ratio, equal to $1/k$
S	Shear strength in the xy or 1–2 plane of a ply
s	A subscript for the shear component in the xy or the 1–2 plane; or ultimate shear strain
X, X'	Longitudinal tensile and compressive strengths
x, x'	Longitudinal tensile and compressive ultimate strains
Y, Y'	Transverse tensile and compressive strengths
y, y'	Transverse tensile and compressive ultimate strains
$\{\epsilon\}, \epsilon_i$	Strain components; $i = x, y, s$
$\{\sigma\}, \sigma_i$	Stress components; $i = x, y, s$
θ_i	Ply orientation

1 QUADRATIC FAILURE CRITERION

It is important to realize that failure criteria are purely empirical. Their purpose is to define a failure envelope by using a minimum number of test data. These data are obtained from relatively simple uniaxial and pure shear tests. Combined stress tests are difficult to perform and are therefore not included in the determination of the failure envelope.

There are, however, geometric and material considerations which will limit the mathematical form of a failure criterion and the shape of the envelope. For example, a failure envelope must be closed in order to prevent infinite strength. Another example is that the envelope must be convex so that unloading from a state of stress will not lead to additional failures.

This article represents the authors' contribution to a world-wide exercise to confirm the state-of-the-art for predicting failure in composites, organized by Hinton and Soden.¹

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A two-dimensional representation of a general quadratic criterion in stress space is shown in the equation below:²³

$$\frac{\sigma_x^2}{XX'} + \frac{2F_{xy}^* \sigma_x \sigma_y}{\sqrt{XX'YY'}} + \frac{\sigma_y^2}{YY'} + \frac{\sigma_s^2}{S^2} + \left[\frac{1}{X} - \frac{1}{X'} \right] \sigma_x + \left[\frac{1}{Y} - \frac{1}{Y'} \right] \sigma_y = 1 \quad (1a)$$

where

$$F_{xy}^* = F_{xy} / \sqrt{F_{xx} F_{yy}} \quad (1b)$$

and, for closed envelopes

$$-1 \leq F_{xy}^* \leq 1 \quad (1c)$$

In eqn (1a) X and X' are the longitudinal tensile and compressive strengths; Y and Y' are the transverse tensile and compressive strengths; S is the longitudinal shear strength; and the normalized interaction term is bounded to ensure that the failure envelope is closed.

The value of this interaction term can best be determined by combined stress tests. In place of performing such difficult tests, it is possible to narrow the bounds between -1 and $+1$ cited in eqn (1a) by considering a more restrictive shape for the envelope. A restriction, for example, can be invoked by the admissible tangents to the envelope at the axis intercepts. This is shown in Figs 8.10 to 8.13 of Ref. 3, where the admissible tangents at X , X' , Y and Y' are defined by shaded areas.

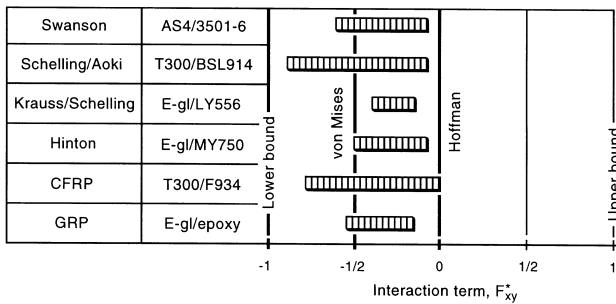


Fig. 1. Range of values for the normalized interaction term for this exercise and for two other common materials, see also Ref. 4.

These constraints are rationalized by virtue of the facts that: (1) there is only one interaction term for the entire stress space, and (2) the envelope must be convex. More restrictive bounds for each material can thus be calculated. The results are also shown as bars in Fig. 1. In this figure, the first four materials are those proposed for this exercise.⁴ The next two are common composite materials, one carbon-fiber-reinforced plastic (CFRP) and one glass-fiber-reinforced plastic (GRP). Note that the values are all zero or negative. In fact, an average value of $F_{xy}^* = -1/2$, referred to as the generalized von Mises model, is reasonable for all materials shown in the Fig. 1.

A comparison with the values of F_{xy}^* in other quadratic failure criteria is given in Table 1. We shall use the generalized von Mises model as the basis of our failure criterion in this exercise. The interaction term for the first two criteria is essentially zero for highly anisotropic materials.

2 STRENGTH RATIOS

In order to facilitate the application of a failure criterion, it is convenient to use either the strength ratio, R , or its reciprocal failure index, k . The strength/stress ratio, or strength ratio for short, is based on proportional loading applied from any state of stress. If loading is from the origin of stress space, the ratio R is easily defined and shown in Fig. 2.

By substituting the maximum stress components into the quadratic failure criterion, we have a quadratic equation in strength ratio R and can easily determine its value. Letting σ_i reach maximum values when

$$F_{ij}\sigma_i^{\max}\sigma_j^{\max} + F_i\sigma_i^{\max} = 1 \quad (2a)$$

we substitute $R\sigma_i^{\text{applied}}$ for $\sigma_i^{\max}$:

$$[F_{ij}\sigma_i\sigma_j]R^2 + [F_i\sigma_i]R - 1 = 0 \quad (2b)$$

Solving the quadratic equation:

$$aR^2 + bR - 1 = 0; \quad a = F_{ij}\sigma_i\sigma_j, b = F_i\sigma_i \quad (2c)$$

Table 1. Interaction terms for various quadratic criteria

Criteria	Uniaxial strengths	F_{xy}	F_{xy}^* (all materials)
Tsai-Hill	$X = X', Y = Y'$	$-\frac{1}{2X^2}$	$-0.014 \leq \frac{Y}{2X} \leq -0.008$
Hoffman	$X \neq X', Y \neq Y'$	$-\frac{1}{2XX'}$	$-0.041 \leq -\frac{1}{2}\sqrt{\frac{YY'}{XX'}} \leq -0.022$
Generalized von Mises	$X \neq X', Y \neq Y'$	$\frac{F_{xy}}{\sqrt{XX'YY'}}$	$-\frac{1}{2}$
Tsai-Wu	$X \neq X', Y \neq Y'$	$\frac{F_{xy}^*}{\sqrt{XX'YY'}}$	$-1 \leq F_{xy}^* \leq 1$

gives the strength ratio R equal to the positive quadratic root

$$R = -(b/2a) + [(b/2a)^2 + 1/a]^{1/2} \quad (2d)$$

This approach is easy to use because the resulting strength ratio provides a linear scaling factor; i.e.

If $R = 1$, failure occurs.

If $R = 2$, the factor of safety is 2. Load can be doubled or laminate thickness reduced by 1/2 before failure occurs.

The same strength ratio can be determined from the equivalent quadratic criterion in strain space. This may be preferred in a laminate because ply strains are either uniform or vary linearly across the thickness. Similarly, the failure index can be determined in the strain space.

R = strength ratio:

$$[G_{ij}\epsilon_i\epsilon_j]R^2 + [G_i\epsilon_i]R = aR^2 + bR = 1 \quad (3)$$

k = failure index:

$$[G_{ij}\epsilon_i\epsilon_j]\left[\frac{1}{k}\right]^2 + [G_i\epsilon_i]\left[\frac{1}{k}\right] = a\left[\frac{1}{k}\right]^2 + \frac{b}{k} = 1$$

3 HYGROTHERMAL STRESSES

Laminates are normally cured at an elevated temperature, and after curing moisture absorption normally occurs. The effects of thermal and moisture stresses from these sources can be readily estimated by using a linear theory of thermoelasticity. These stresses and strains can significantly affect the first-ply-failure condition. Since many designs are based on this approach, these hygrothermal stresses should not be ignored.

It is therefore recommended that the effects of both elevated-temperature curing and moisture absorption be included in the strength prediction of a laminate. The expansion coefficients used for our calculation are taken from Ref. 3, Table 4.4, and Ref. 4 where

- Swanson material is based on T300/5208 in Table 4.4;
- Schelling material is based on AS/3501 in Table 4.4;
- Krauss and Hinton materials are based on E-glass/epoxy in Table 4.4.

We have assumed that the temperature difference is -100°C and that the moisture content is 0.005 or 0.5%. These two factors tend to cancel each other out if both occur. If a laminate is cured at room temperature, then moisture absorption can have a significant effect on ply failures. Conversely, if a laminate is cured at elevated temperature but used in a dry environment, then curing stresses will not be cancelled and can be significant.

4 MICRO-CRACKING

A ply in a laminate may fail by micro-cracking when the transverse strain on the ply axis is positive (tensile). A saturation level of periodically dispersed cracks is reached. This phenomenon will be explained below. When the transverse strain component is zero or negative the failure mechanism will not be micro-cracking. It would fail by crushing or buckling. Since transverse compressive failure strain, y' , shown in Table 2 is many times higher than the transverse tensile strain, y , or even higher than the longitudinal compressive failure strain, x' , we assume that the ply failure will be a combined matrix and fiber failure. The failure will occur in one location, totally different from the widely dispersed micro-cracks.

If micro-cracking takes place in a ply, we assume that it happens instantaneously within a limited region of high stress in a ply. Having other plies at the same point in a laminate, the laminate as a whole may be capable of continuing to carry the prevailing load. It is therefore useful to model the presence of micro-cracking in a ply by reducing its transverse and shear moduli while maintaining the longitudinal stiffness. The extent of this reduction is a function of the number of cracks at saturation as measured by their density or aspect ratio. A shear lag model for a [0/90] laminate of T300/5208 CFRP is shown in Fig. 3, where the reduction in stiffness of the laminate approaches 5% as micro-cracks reach a saturation level.⁵

Although reductions in transverse and shear moduli at the crack saturation level can be calculated for each laminate subjected to a uniaxial tensile load, a simpler method of estimating this reduction can be obtained in terms of a matrix reduction factor by using

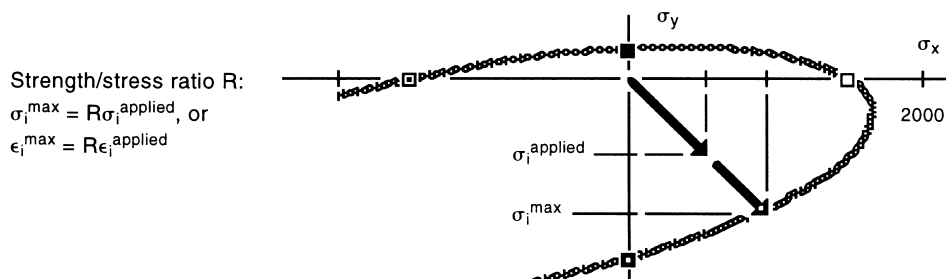


Fig. 2. Strength ratio R as the scaling factor of a loading vector.

micromechanics.³⁵ For the purpose of this exercise, we only list the modulus reduction in both absolute and normalized forms (with respect to those of the intact plies and shown with asterisks) in Table 3. The estimated fiber volume fractions are also listed.

5 LONGITUDINAL COMPRESSIVE STRENGTH

Another factor that is affected by the presence of micro-cracking is the longitudinal compressive strength of a ply within a laminate. Using the concept of a

beam on an elastic foundation, we can show the loss in the compressive strength by the reduction in the longitudinal shear modulus. This approach was used by Rosen⁶ for the loss in compressive strength by in-phase fiber buckling (symmetric).

The predicted loss is linearly proportional to the reduction in shear modulus. The loss, however, seems more severe than necessary. We therefore introduce an exponent n to vary the loss. When $n = 1$ we have Rosen's linear model. When $n < 1$ we can dampen the effect of the loss in compressive strength. We recommend the use of $n = 0.1$ which lies between the values $n = 0.0$ and $n = 0.2$ shown in Fig. 4.

Table 2. Ultimate stress and strain of the exercise and other materials⁴

	Swanson	Schelling	Krauss	Hinton	CFRP	GRP
Fiber	AS4	T300	E-glass	E-glass	T300	E-glass
Matrix	3501-6	BSL914C	LY556	MY750	F934	epoxy
<i>Max stress (MPa)</i>						
X	1950	1500	1140	1280	1314	1062
X'	1480	900	570	800	1220	610
Y	48	27	35	40	43	31
Y'	200	200	114	145	168	118
S	79	80	72	73	48	72
<i>Max strain, eps* ($\times 10^{-3}$)</i>						
x	151.37	10.80	20.77	27.30	8.88	27.51
x'	11.67	6.48	10.39	17.06	8.24	15.80
y	4.33	2.44	1.93	2.40	4.46	3.75
y'	18.06	18.07	6.28	8.70	17.41	14.27
s	11.97	14.55	12.35	12.52	10.55	17.39

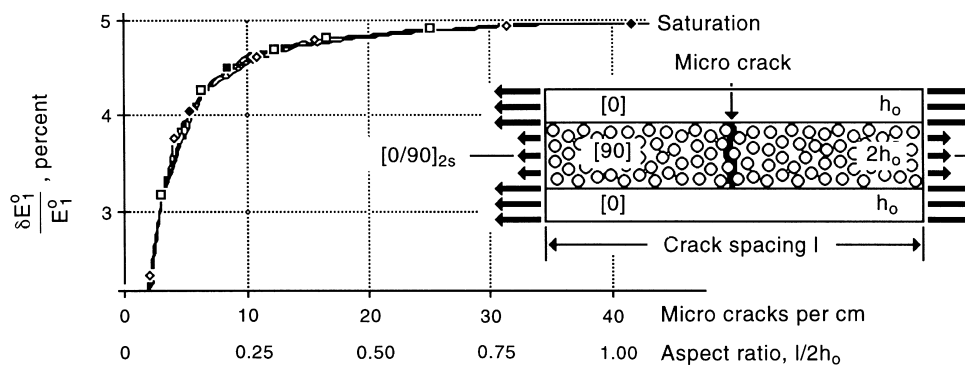


Fig. 3. Asymptotic loss in laminate stiffness based on shear lag analysis as micro-cracking reaches a saturation level.

Table 3. Reduced and transverse shear moduli of plies due to micro-cracks

Degraded moduli GPa/normalized		Swanson AS4/35	Schelling T300/9	Krauss E-gl/ep	Hinton E-gl/ep	CFRP T300/F9	GRP E-gl/ep
Matrix	E_m	→ 0.51	→ 0.51	→ 0.51	→ 0.51	→ 0.51	→ 0.51
	E_m^*	→ 0.15	→ 0.15	→ 0.15	→ 0.15	→ 0.15	→ 0.15
Transverse	E_y	1.93	1.93	2.19	1.71	1.93	1.32
	E_y^*	0.18	0.18	0.12	0.11	0.20	0.16
Shear	E_s	1.07	1.04	1.11	0.86	1.00	0.66
	E_s^*	0.16	0.19	0.19	0.15	0.22	0.16
Fiber volume fraction		0.60	0.60	0.62	0.60	0.60	0.45

6 SUMMARY OF DEGRADATION FACTORS

Subjective derived factors of degradation as absolute and normalized (modified/baseline) values are tabulated in Table 4 where:

- matrix degradation factor, $E_m^* = 0.15$ (Table 3);
- fiber degradation, $E_f^* = 0.01$, signifying a catastrophic collapse;
- compressive strength degradation exponent, $n = 0.1$.

The composite ply material used by Swanson in this exercise will sustain the degradation indicated in Table 4.

We have made many studies of the sensitivities of various degradation factors to the resulting final (LPF)

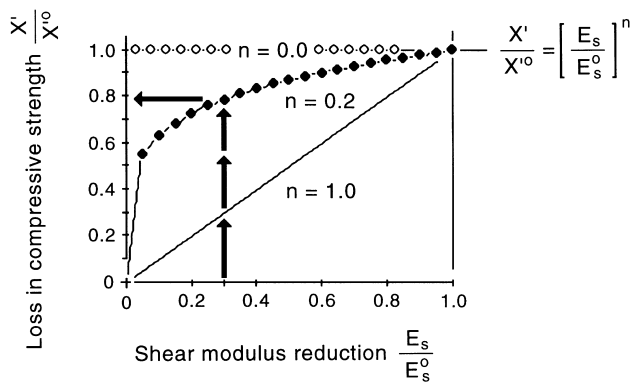


Fig. 4. An estimated reduction in compressive strength due to the reduction in shear modulus caused by micro-cracking.

failure envelopes and found that there are so many uncertainties that the exact numerical values will not significantly affect the qualitative nature of the resulting predictions. For example, the selective degradation (as determined by the sign of the transverse ply strain) can have a greater influence than the values of the degradation factors.

7 PROGRESSIVE FAILURE SCENARIO

With the degradation factors, we can now outline a progressive failure scenario of a laminate subjected to applied stresses. First of all, the traditional application of a failure criterion is shown in the box on the left-hand side of Fig. 5. Ply material and ply orientation are selected to form a laminate. After a load $\{N\}$ is applied, the first ply failure (FPF) is determined based on the ply having the minimum or lowest strength ratio among all plies.

Secondly, the transverse strain of the failed ply will dictate whether micro-cracking has taken place. Only when this strain component is positive will micro-cracking occur. The ply will be degraded by a reduced matrix modulus, say to 0.15 of the virgin modulus or the values of transverse and shear moduli listed in Table 3. If the transverse strain is zero or compressive, the ply is assumed to remain intact (no micro-cracking) and the only possible failure mode would be that of the fiber. This failure mode is catastrophic as shown by the

Table 4. A summary of stiffness and strength of virgin and degraded AS4/3501-6 ply material (Swanson)

	Intact	Matrix degradation		Fiber degradation	
	Baseline	Modified	Mod/B	Modified	Mod/B
E_f^*	210	210	1.000	2.100	$0.010 \leftarrow E_f^*$
E_m^*	3.40	0.51	$0.150 \leftarrow E_m^*$	0.034	$0.010 \leftarrow E_f^*$
E_x	126.00	126.00	1.000	1.260	$0.010 \leftarrow E_f^*$
E_y	11.00	1.930	0.175	0.110	$0.010 \leftarrow E_f^*$
ν_x	0.28	0.042	$0.150 \leftarrow E_m^*$	0.003	$0.010 \leftarrow E_f^*$
E_s	6.60	1.070	0.162	0.066	$0.010 \leftarrow E_f^*$
X	1950	1950	1.00	1950	1.00
X'	1480	1258	$0.85 \leftarrow n=0.1$	976	$0.66 \leftarrow n=0.1$
Y	48	48	1.00	48	1.00
Y'	200	200	1.00	200	1.00
S	79	79	1.00	79	1.00
F_{xy}	-0.50	-0.08	$0.15 \leftarrow E_m^*$	-0.01	$0.01 \leftarrow E_f^*$

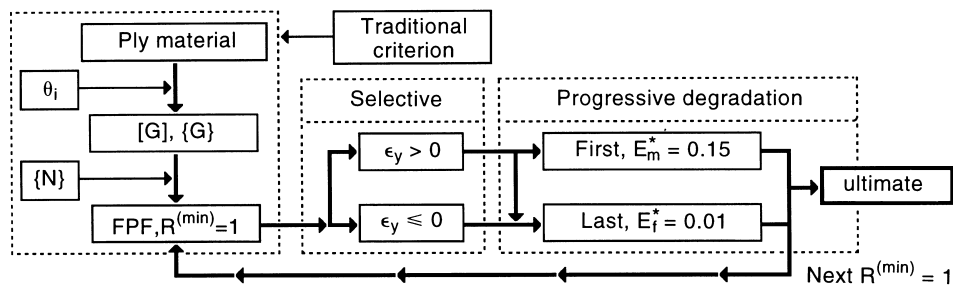


Fig. 5. A flowchart of the progressive failure modeling that extends the traditional FPF to include matrix and fiber failure modes.

application of an arbitrary longitudinal stiffness degradation factor of 0.01, or 1% of the virgin stiffness.

Thirdly, having a reduced matrix or fiber modulus, the next ply failure can be calculated. The failed ply will again be selectively degraded depending on whether or not the transverse strain is positive. If this second ply is the same ply that has failed by matrix degradation

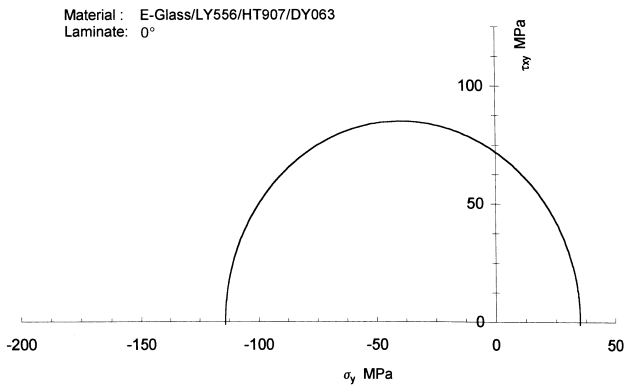


Fig. 6. Biaxial failure stress envelope for 0° unidirectional lamina made of glass/epoxy composite under transverse and shear loading (σ_y versus τ_{xy}).

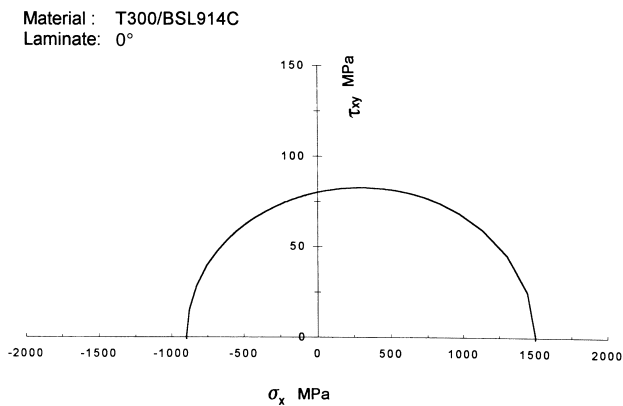


Fig. 7. Biaxial failure stress envelope for 0° unidirectional lamina made of T300/BSL914C composite under longitudinal and shear loading (σ_x versus τ_{xy}).

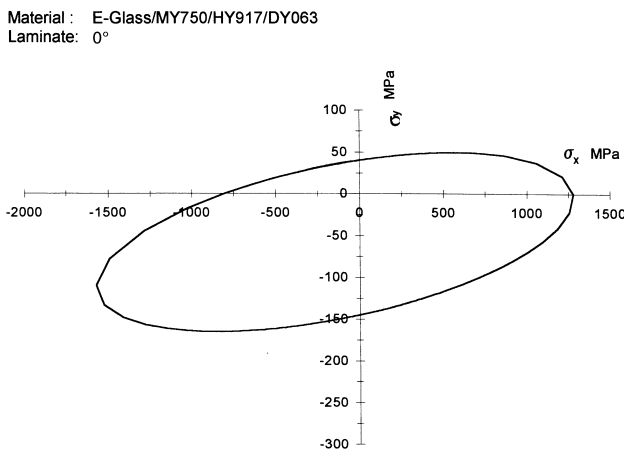


Fig. 8. Biaxial failure stress envelope for 0° unidirectional lamina made of glass/epoxy composite under longitudinal and transverse loading (σ_y versus σ_x).

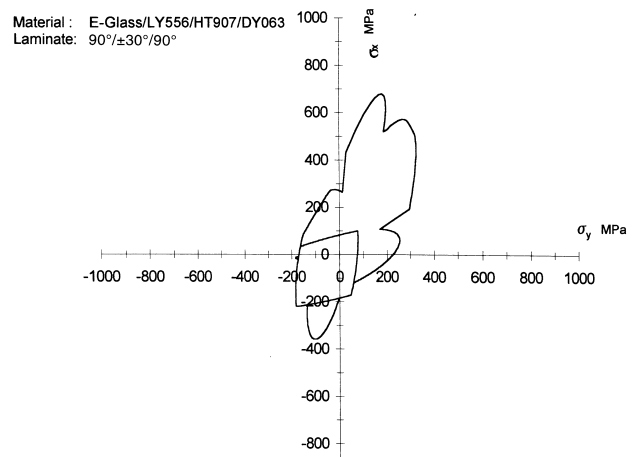


Fig. 9. Biaxial failure stress envelope for $(90^\circ/\pm 30^\circ/90^\circ)$ laminate made of glass/epoxy composite under combined loading (σ_y versus σ_x).

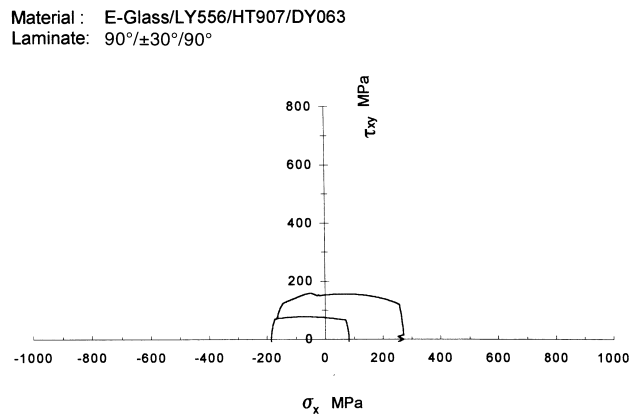


Fig. 10. Biaxial failure stress envelope for $(90^\circ/\pm 30^\circ/90^\circ)$ laminate made of glass/epoxy composite under combined loading (σ_x versus τ_{xy}).

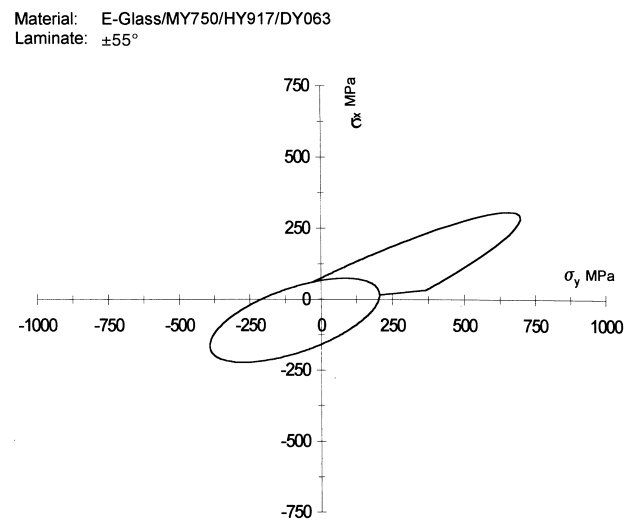


Fig. 11. Biaxial failure stress envelope for $\pm 55^\circ$ angle-ply laminate made of glass/epoxy composite under combined loading (σ_y versus σ_x).

(from micro-cracking), fiber degradation will be the only path. Thus, this ply will have to be degraded twice. But if the first ply has failed as a result of fiber failure, it cannot fail again through matrix failure. This ply can only fail once.

Finally, the process of progressive failures on a ply-by-ply basis will continue until the maximum load is reached, beyond which the load will reduce as additional plies fail. The ultimate load of the laminate is thus determined.

8 COMMENTS ON THE BASELINE DATA PROVIDED

Details of the lamina properties, lay-up configurations and loading of the laminates analysed are provided in Ref. 4.

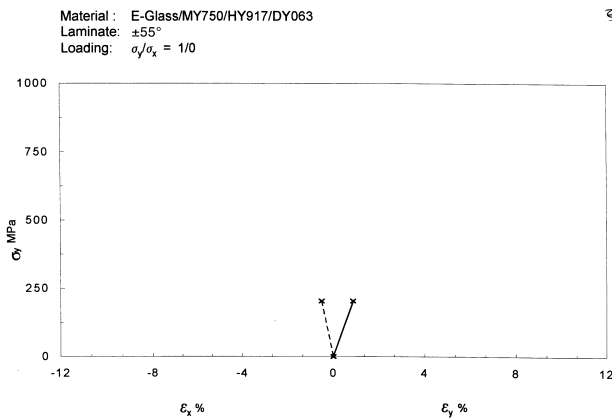


Fig. 12. Stress/strain curves for $\pm 55^\circ$ angle-ply laminate made of glass/epoxy composite under uniaxial tensile loading with $\sigma_y/\sigma_x = 1/0$.

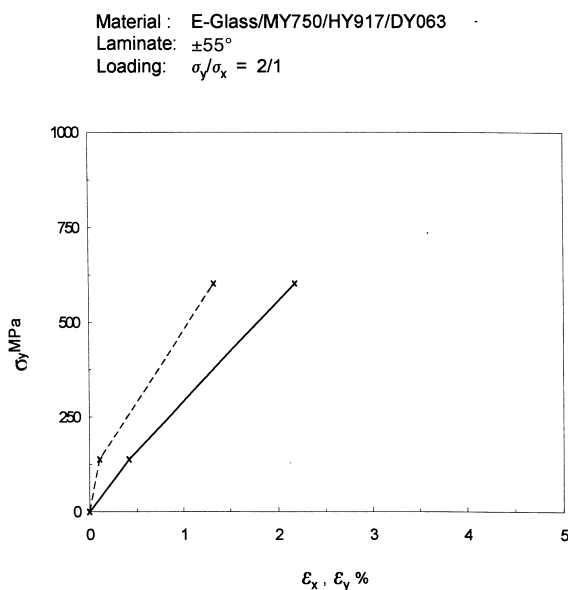


Fig. 13. Stress/strain curves for $\pm 55^\circ$ angle-ply laminate made of glass/epoxy composite under biaxial tensile loading with $\sigma_y/\sigma_x = 2/1$.

In the prediction of failure envelopes and stress/strain curves using our progressive failure model, our analysis is linear. For materials that exhibit non-linear stress/strain curves, we used the secant modulus as the linear modulus from the origin to failure. The failure strain is determined by the ratio of ultimate strength to the secant modulus.

All other properties given by the editor of this exercise were used for the predicted results shown in all subsequent figures (Figs 6–19), with the following additional properties.

- The degradation factors, however, were selected by us and are all listed in Table 3 for the transverse and shear moduli.
- Additional properties are the fiber degradation and compressive strength exponent shown in Table 4.

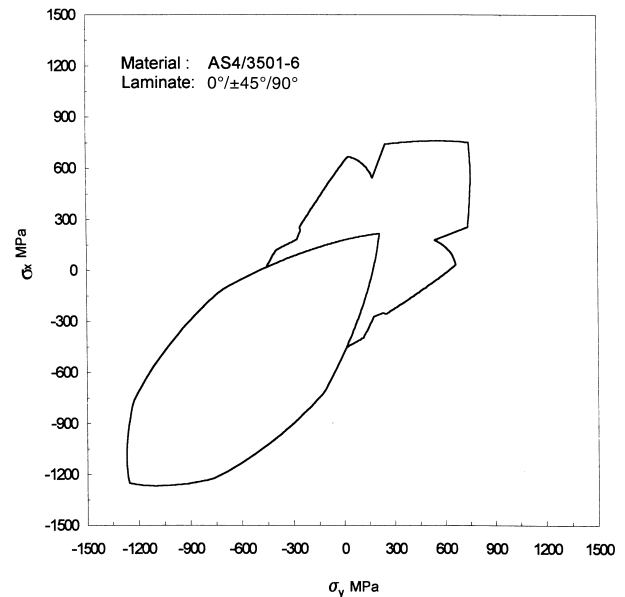


Fig. 14. Biaxial failure stress envelope for $(0^\circ/\pm 45^\circ/90^\circ)$ laminate made of AS4/3501-6 composite under combined loading (σ_x versus σ_y).

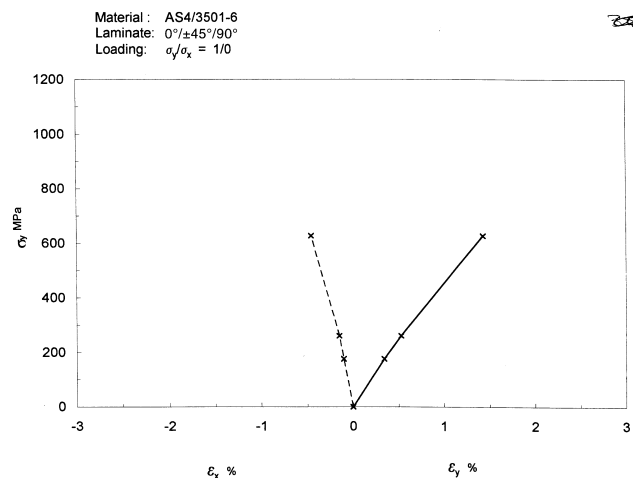


Fig. 15. Stress/strain curves for $(0^\circ/\pm 45^\circ/90^\circ)$ laminate made of AS4/3501-6 composite under uniaxial tensile loading in y direction ($\sigma_y/\sigma_x = 1/0$).

- The hygrothermal constants used for curing stresses and moisture absorption follow those recommended in Ref. 3, Table 4.4. These values differ from those provided by the organizers of this study, but the effect on the resulting hygrothermal stresses is not significant.

9 OBSERVATIONS ON PREDICTED RESULTS

Figures 6–8 show the failure envelopes of the UD laminae. The first and last failure are the same in the Figs 9–11 and Fig. 14 where more than one failure envelope has been plotted, the envelope that contains the origin will

be a first-ply-failure (FPF) envelope within which no failure occurs. Envelopes beyond the FPF are the final failure or the last-ply-failure (LPF). Degradation occurs instantaneously when FPF is reached. If proportional loading continues to increase beyond FPF, progressive ply failures occur until the LPF is reached. The envelopes plotted show only FPF and LPF and do not show the intermediate plies.

An LPF envelope is not smooth because ply orientations vary depending on the combined stresses applied. The convexity of the envelopes is not violated if radial loading lines are drawn from the origins. At certain loading vectors, the predicted LPF jumps from one ply orientation to another. The jump gives the appearance of non-convexity.

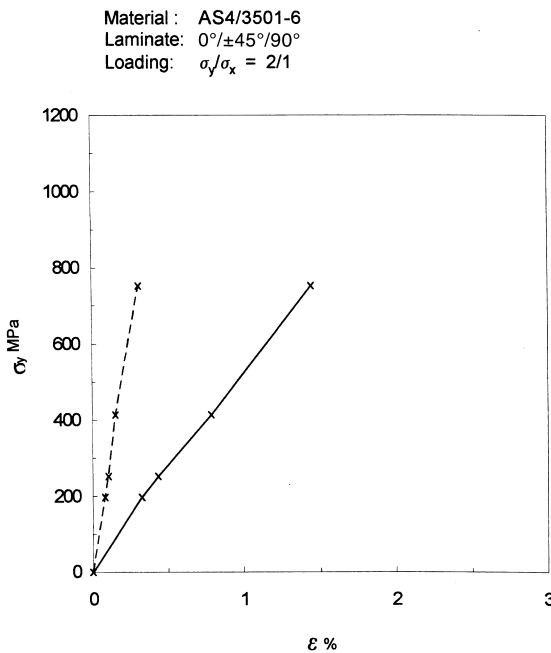


Fig. 16. Stress/strain curves for $(0^\circ/\pm 45^\circ/90^\circ)$ laminate made of AS4/3501-6 composite under biaxial tensile loading with $\sigma_y/\sigma_x = 2/1$.

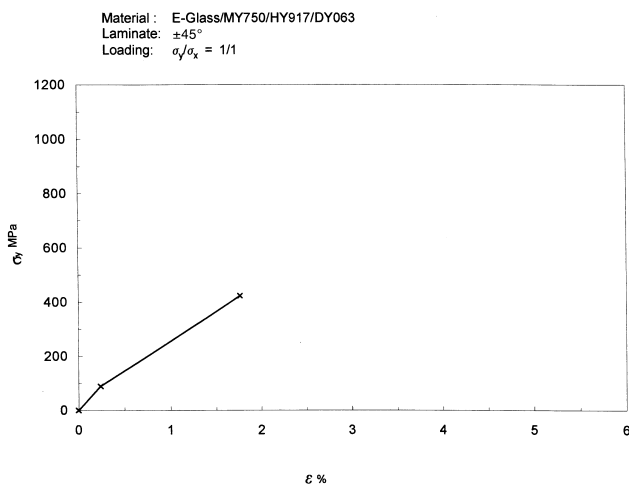


Fig. 17. Stress/strain curves for $\pm 45^\circ$ angle-ply laminate made of glass/epoxy composite under biaxial tensile loading with $\sigma_y/\sigma_x = 1/1$.

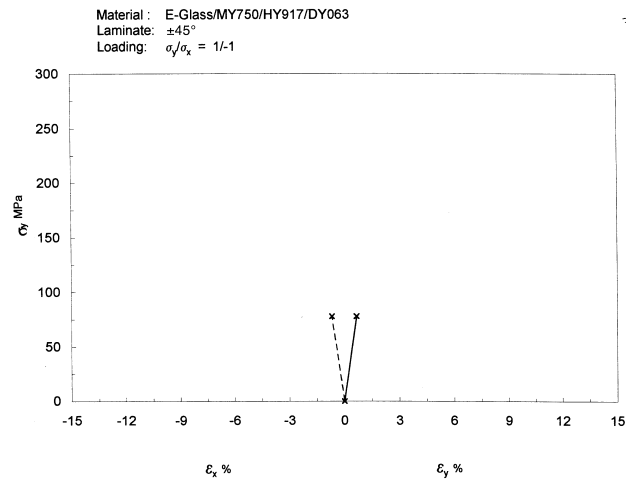


Fig. 18. Stress/strain curves for $\pm 45^\circ$ angle-ply laminate made of glass/epoxy composite under biaxial tensile loading with $\sigma_y/\sigma_x = 1/-1$.

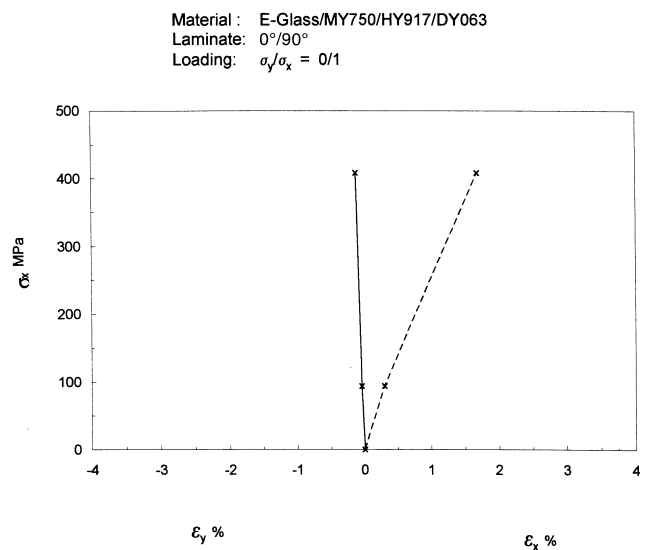


Fig. 19. Stress/strain curves for $(0^\circ/90^\circ)$ cross-ply laminate made of glass/epoxy composite under uniaxial tensile loading with $\sigma_y/\sigma_x = 0/1$.

In Figs 15 and 16, where uniaxial and 2:1 biaxial stresses are imposed on $[\pi/4]$ laminates, the successive failure plies are shown by 'x' on the stress strain curves. The specific ply that is associated with the 'x' is as follows:

- For uniaxial tensile loading, shown in Fig. 15, the lowest 'x' was $[90]$, followed by $[\pm 45]$, and ultimately $[0]$. Each ply failed only once; i.e. each went from intact to degraded by micro-cracking. Thus when the ultimate stress was reached, all plies were saturated with micro-cracks.
- For biaxial tensile loading, the lowest 'x' was $[90]$, followed by $[\pm 45]$, and then $[0]$ micro-cracked, and ultimately $[0]$ collapsed. This progression is different from the uniaxial loading by having the $[0]$ ply fail twice.

This failure scenario is explained in Fig. 5. In uniaxial loading, after all plies including $[0]$ had micro-cracks, the degraded laminate could not carry any more load. The ultimate stress was thus reached. For biaxial loading, the $[0]$ ply could continue to carry a load after micro-cracking. Thus fiber failure must set in before the ultimate stress is reached.

In Fig. 13, where a $[\pm 55]$ laminate is subjected to hydrostatic stress, the lower 'x' signified the FPF or formation of micro cracks, and the higher 'x' the fiber failures at LPF. Thus each ply went through two failures, similar to the $[0]$ ply of $[\pi/4]$ subjected to 2:1 stress above.

In Fig. 18, where the $[\pm 45]$ lay-up is subjected to tension and compression, FPF coincides with LPF, implying that after micro-cracks set in the laminate could not carry any more load. The solid line represents strain in the y direction and the dashed line represents that in the x direction. The behavior is the same as imposing pure shear on $[0/90]$, in which case there is no post-FPF load-carrying capability.

10 CONCLUDING REMARKS

The predictions of failure envelopes and stress/strain curves for various composite laminates are attached. The failures of laminates on a progressive basis can be found in Tsai's work.³ Again we recognize that failure criteria are empirical. The progressive failure scenario described here can be applied to other failure criteria. Our favorite is the quadratic criterion for its accuracy, ease of use, flexibility, scalar representation, and so on. A criterion is only as good as the data available. More data for wide-ranging combinations of stresses seem to agree with the quadratic criterion than with any other criterion. It is also important to recognize that a failure criterion is often the only basis for interpolating and extrapolating strength under combined stresses where data are not available.

Micro-cracking has been a mechanism of failure extensively studied in many investigations. This

mechanism occurs under transverse tensile/longitudinal shear loading conditions. We use a selective degradation criterion to distinguish this from non-tensile conditions. Compression and shear failures are caused by crushing and buckling, and involve no micro-cracking. Then a ply is considered to have failed totally, as implemented by a fiber collapse. The FPF represents the ultimate load and is the one and only LPF.

Our criterion is intended to describe intralaminar failures, in 1, 2 and 3 dimensions. It can be extended to interlaminar failures if normal and tangential strengths can be measured.

From our experience with many existing composite materials, we conclude that progressive failure of a point-stress analysis is not as important as that of a structural analysis having non-homogeneous stresses. A simple example would be the failure progression of a plate with an open hole subjected to a uniaxial tensile load. There will be a stress concentration, and ply failures will extend from the free edge of the hole into the interior of the laminate. Thus at each point within the laminate ply-by-ply failure may be modeled by the progressive model described in this paper. Then a ply failure can extend into neighboring points as the applied load increases. The process will continue until the applied load reaches a maximum, after which the load will drop as failure expands within plies and among points or elements. It will take a sophisticated finite-element analysis to track the progressive failures of a laminate under non-homogeneous stresses. But the scenario described here may provide a good starting point.

We are not happy with the degree of empiricism that we have used in the progressive failure scenario. First of all, the failure process is extremely complicated. Our attempt is a much simplified approach. What we have not studied includes delamination and the limitations of the homogenization of micro-cracking. While new approaches will always be welcome, the best justification of failure criteria can be made from the standpoint of convenience and utility.

The present exercise is an effective way to address this important subject. A follow-up to verify non-homogeneous stresses may be equally important. Test data on open-hole and plugged-hole tensile and compressive loading conditions are available and can serve as starting points for the next exercise.

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